ASSIGNMENT: 11

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Subject: Advanced Programming

```
from abc import ABC, abstractmethod

# Abstract Base Class
class LibraryItem(ABC):
    item_count = 0  # class-level counter (extra feature)

    def __init__(self, title, year):
        self.title = title
        self.year = year
        LibraryItem.item_count += 1

    @abstractmethod
    def displayInfo(self):
        pass

    @classmethod
    def get_item_count(cls):
        return cls.item_count


# Book Class
class Book(LibraryItem):
    def __init__(self, title, year, author="Unknown"):
        super().__init__(title, year)
        self.author = author

    # Method overriding
    def displayInfo(self):
        print(f"[Book] Title: {self.title}")
        print(f"Year: {self.year}")
        print(f"Author: {self.author}")


# DVD Class
class DVD(LibraryItem):
    def __init__(self, title, year, duration=0, genre="Unknown"):
        super().__init__(title, year)
```

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        self.duration = duration  # in minutes
        self.genre = genre

    # Method overriding
    def displayInfo(self):
        print(f"[DVD] Title: {self.title}")
        print(f"Year: {self.year}")
        print(f"Duration: {self.duration} mins")
        print(f"Genre: {self.genre}")


# Main Execution
if __name__ == "__main__":
    # Polymorphism: storing different objects in same list
    items = [
        Book("Python Basics", 2022, "John Doe"),
        DVD("Inception", 2010, 148, "Sci-Fi"),
        Book("Data Structures", 2021),   # uses default author
        DVD("Nature Documentary", 2018)  # uses default values
    ]

    for item in items:
        print("\n--- Library Item ---")
        item.displayInfo()  # polymorphic call

    print("\nTotal Items in Library:", LibraryItem.get_item_count())
```